

UNIT-6 PROBABILITY

CASE BASED QUESTIONS

- 1.** A departmental store sends bills to charge its customers once a month. Past experience shows that 70% of its customers pay their first month bill in time. The store also found that the customer who pays the bill in time has the probability of 0.8 of paying in time next month and the customer who doesn't pay in time has the probability of 0.4 of paying in time the next month.
- Based on the above information, answer the following questions :
- (i) Let E_1 and E_2 respectively denote the event of customer paying or not paying the first month bill in time.
Find $P(E_1)$, $P(E_2)$.
 - (ii) Let A denotes the event of customer paying second month's bill in time, then find $P(A|E_1)$ and $P(A|E_2)$.
 - (iii) Find the probability of customer paying second month's bill in time.
- OR**
- (iii) Find the probability of customer paying first month's bill in time if it is found that customer has paid the second month's bill in time.

2. Rohit, Jaspreet and Alia appeared for an interview for three vacancies in the same post. The probability of Rohit's selection is $\frac{1}{5}$, Jaspreet's selection is $\frac{1}{3}$ and Alia's selection is $\frac{1}{4}$. The event of selection is independent of each other.



Based on the above information, answer the following questions :

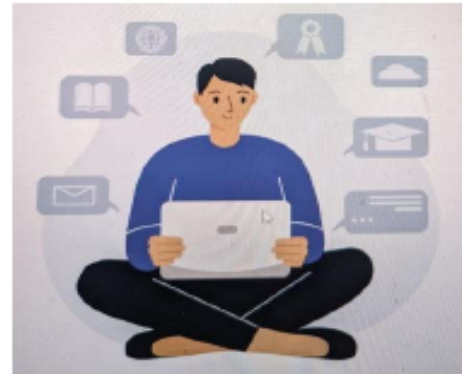
- (i) What is the probability that at least one of them is selected ?
- (ii) Find $P(G | \bar{H})$ where G is the event of Jaspreet's selection and $\bar{H}$ denotes the event that Rohit is not selected.
- (iii) Find the probability that exactly one of them is selected.

OR

- (iii) Find the probability that exactly two of them are selected.

3.

Self-study helps students to build confidence in learning. It boosts the self-esteem of the learners. Recent surveys suggested that close to 50% learners were self-taught using internet resources and upskilled themselves.



A student may spend 1 hour to 6 hours in a day in upskilling self. The probability distribution of the number of hours spent by a student is given below :

$$P(X = x) = \begin{cases} kx^2, & \text{for } x = 1, 2, 3 \\ 2kx, & \text{for } x = 4, 5, 6 \\ 0, & \text{otherwise} \end{cases}$$

where x denotes the number of hours.

Based on the above information, answer the following questions :

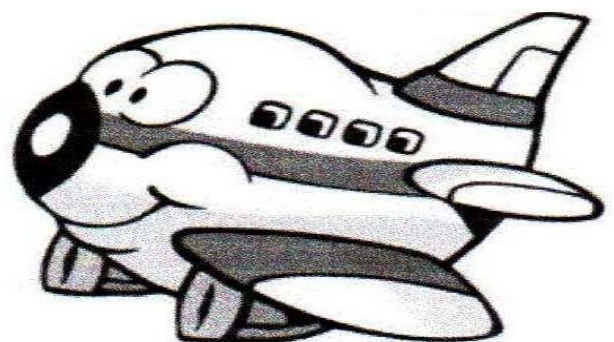
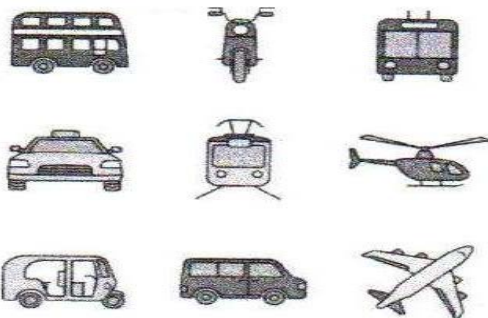
- (i) Express the probability distribution given above in the form of a probability distribution table.
- (ii) Find the value of k .
- (iii) (a) Find the mean number of hours spent by the student.

OR

- (iii) (b) Find $P(1 < X < 6)$.

4.

Airplanes are by far the safest mode of transportation when the number of transported passengers are measured against personal injuries and fatality totals.



Previous records state that the probability of an airplane crash is 0.00001%. Further, there are 95% chances that there will be survivors after a plane crash. Assume that in case of no crash, all travellers survive.

Let E_1 be the event that there is a plane crash and E_2 be the event that there is no crash. Let A be the event that passengers survive after the journey.

On the basis of the above information, answer the following questions :

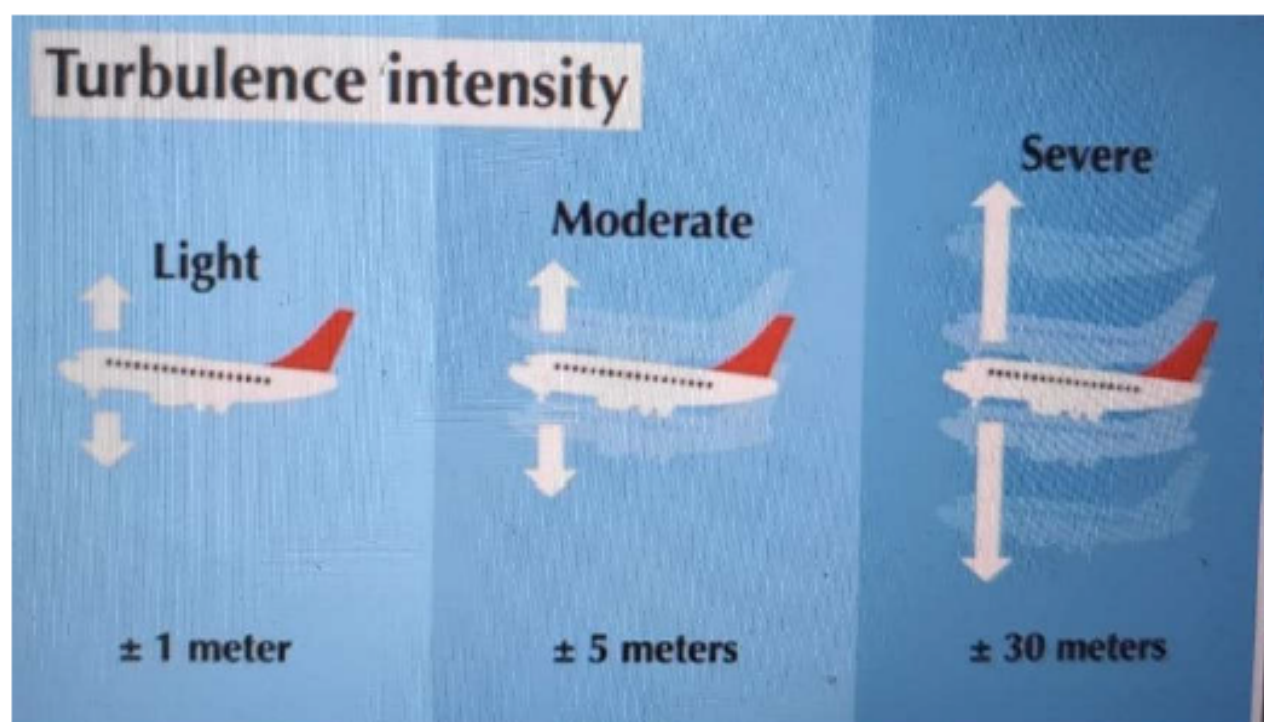
- (i) Find the probability that the airplane will not crash.
- (ii) Find $P(A | E_1) + P(A | E_2)$.
- (iii) (a) Find $P(A)$.

OR

- (iii) (b) Find $P(E_2 | A)$.

5. According to recent research, air turbulence has increased in various regions around the world due to climate change. Turbulence makes flights bumpy and often delays the flights.

Assume that, an airplane observes severe turbulence, moderate turbulence or light turbulence with equal probabilities. Further, the chance of an airplane reaching late to the destination are 55%, 37% and 17% due to severe, moderate and light turbulence respectively.



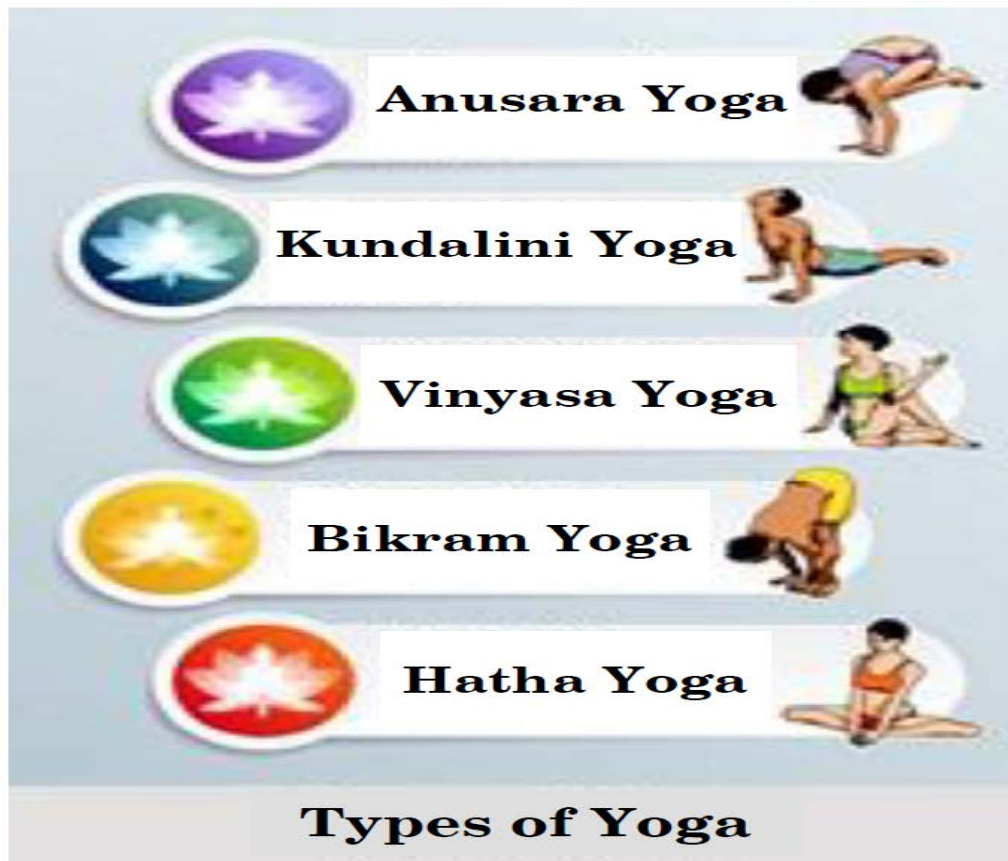
On the basis of the above information, answer the following questions :

- (i) Find the probability that an airplane reached its destination late. 2
- (ii) If the airplane reached its destination late, find the probability that it was due to moderate turbulence. 2

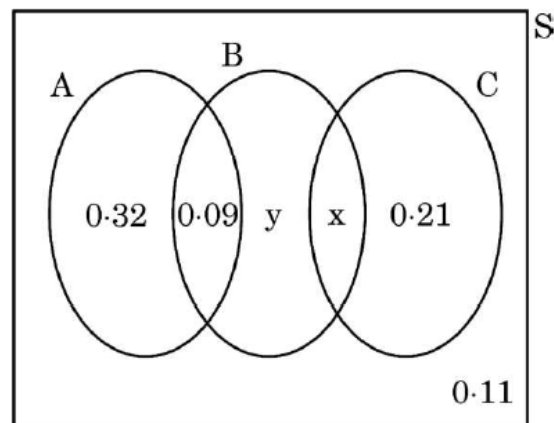
6.

Case Study – 1

There are different types of Yoga which involve the usage of different poses of Yoga Asanas, Meditation and Pranayam as shown in the figure below :



The Venn diagram below represents the probabilities of three different types of Yoga, A, B and C performed by the people of a society. Further, it is given that probability of a member performing type C Yoga is 0.44.



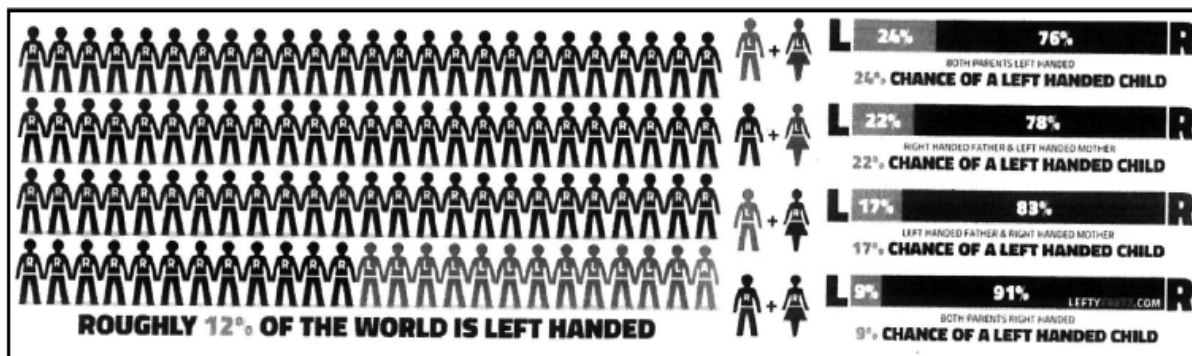
On the basis of the above information, answer the following questions :

- (i) Find the value of x. 1
- (ii) Find the value of y. 1
- (iii) (a) Find $P\left(\frac{C}{B}\right)$. 2

OR

- (iii) (b) Find the probability that a randomly selected person of the society does Yoga of type A or B but not C. 2

7. Recent studies suggest that roughly 12% of the world population is left handed.



Depending upon the parents, the chances of having a left handed child are as follows :

- A : When both father and mother are left handed :
Chances of left handed child is 24%.
- B : When father is right handed and mother is left handed :
Chances of left handed child is 22%.
- C : When father is left handed and mother is right handed :
Chances of left handed child is 17%.
- D : When both father and mother are right handed :
Chances of left handed child is 9%.

Assuming that $P(A) = P(B) = P(C) = P(D) = \frac{1}{4}$ and L denotes the event that child is left handed.

Based on the above information, answer the following questions :

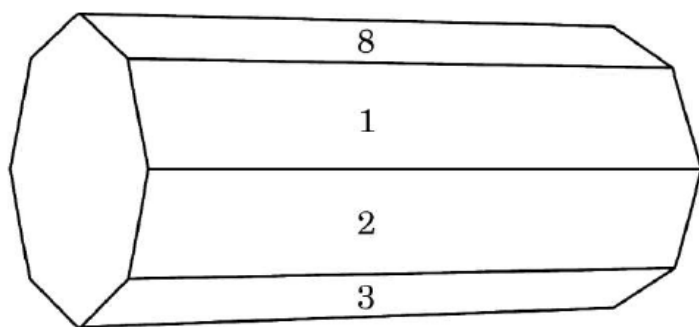
- Find $P(L/C)$
- Find $P(\bar{L}/A)$
- (a) Find $P(A/L)$

OR

- Find the probability that a randomly selected child is left handed given that exactly one of the parents is left handed.

8.

An octagonal prism is a three-dimensional polyhedron bounded by two octagonal bases and eight rectangular side faces. It has 24 edges and 16 vertices.



The prism is rolled along the rectangular faces and number on the bottom face (touching the ground) is noted. Let X denote the number obtained on the bottom face and the following table give the probability distribution of X .

$X :$	1	2	3	4	5	6	7	8
$P(X) :$	p	$2p$	$2p$	p	$2p$	p^2	$2p^2$	$7p^2 + p$

Based on the above information, answer the following questions :

- (i) Find the value of p .
- (ii) Find $P(X > 6)$.
- (iii) (a) Find $P(X = 3m)$, where m is a natural number.

OR

- (iii) (b) Find the mean $E(X)$.

9. A building contractor undertakes a job to construct 4 flats on a plot along with parking area. Due to strike the probability of many construction workers not being present for the job is 0.65. The probability that many are not present and still the work gets completed on time is 0.35. The probability that work will be completed on time when all workers are present is 0.80.

Let : E_1 : represent the event when many workers were not present for the job;

E_2 : represent the event when all workers were present; and

E : represent completing the construction work on time.

Based on the above information, answer the following questions :

- (i) What is the probability that all the workers are present for the job ?
- (ii) What is the probability that construction will be completed on time ?
- (iii) (a) What is the probability that many workers are not present given that the construction work is completed on time ?

OR

- (iii) (b) What is the probability that all workers were present given that the construction job was completed on time ?

10.

A salesman receives a commission for each sale he makes together with a fixed daily income. The number of sales he makes in a day along with their probabilities are given in the table below :

X :	0	1	2	3	4	5
P(X) :	0.42	3k	0.3	0.05	2k	0.03



His daily income Y (in ₹) is given by :

$$Y = 800X + 50$$

On the basis of the above information, answer the following questions :

- (i) Find the value of k .
- (ii) Evaluate $P(X \geq 3)$.
- (iii) (a) Calculate the expected weekly income of the salesman assuming he works five days per week.

OR

- (iii) (b) Calculate the expected weekly income of the salesman assuming he works only for three days of the week.

- 11.** In a group activity class, there are 10 students whose ages are 16, 17, 15, 14, 19, 17, 16, 19, 16 and 15 years. One student is selected at random such that each has equal chance of being chosen and age of the student is recorded.



On the basis of the above information, answer the following questions :

- (i) Find the probability that the age of the selected student is a composite number.
- (ii) Let X be the age of the selected student. What can be the value of X ?
- (iii) (a) Find the probability distribution of random variable X and hence find the mean age.

OR

- (iii) (b) A student was selected at random and his age was found to be greater than 15 years. Find the probability that his age is a prime number.

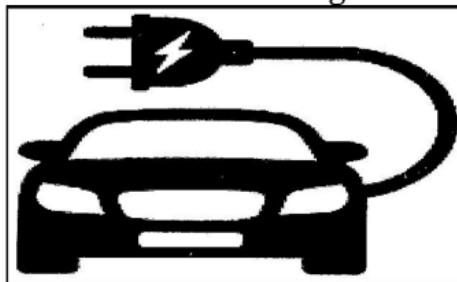
- 12.** A bank offers loan to its customers on different types of interest namely, fixed rate, floating rate and variable rate. From the past data with the bank, it is known that a customer avails loan on fixed rate, floating rate or variable rate with probabilities 10%, 20% and 70% respectively. A customer after availing loan can pay the loan or default on loan repayment. The bank data suggests that the probability that a person defaults on loan after availing it at fixed rate, floating rate and variable rate is 5%, 3% and 1% respectively.



Based on the above information, answer the following :

- (i) What is the probability that a customer after availing the loan will default on the loan repayment ? 2
- (ii) A customer after availing the loan, defaults on loan repayment. What is the probability that he availed the loan at a variable rate of interest ? 2

- 13.** Three persons viz. Amber, Bonzi and Comet are manufacturing cars which run on petrol and on battery as well. Their production share in the market is 60%, 30% and 10% respectively. Of their respective production capacities, 20%, 10% and 5% cars respectively are electric (or battery operated). Based on the above, answer the following :



	(i) (a) What is the probability that a randomly selected car is an electric car ? 2 OR (i) (b) What is the probability that a randomly selected car is a petrol car ? 2 (ii) A car is selected at random and is found to be electric. What is the probability that it was manufactured by Comet ? 1 (iii) A car is selected at random and is found to be electric. What is the probability that it was manufactured by Amber or Bonzi ? 1												
14.	Some students are having a misconception while comparing decimals. For example, a student may mention that $78.56 > 78.9$ as $7856 > 789$. In order to assess this concept, a decimal comparison test was administered to the students of class VI through the following question : In the recently held Sports Day in the school, 5 students participated in a javelin throw competition. The distances to which they have thrown the javelin are shown below in the table : <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <th style="width: 40%;">Name of student</th><th style="width: 60%;">Distance of javelin (in meters)</th></tr> <tr> <td>Ajay</td><td>47.7</td></tr> <tr> <td>Bijoy</td><td>47.07</td></tr> <tr> <td>Kartik</td><td>43.09</td></tr> <tr> <td>Dinesh</td><td>43.9</td></tr> <tr> <td>Devesh</td><td>45.2</td></tr> </table> The students were asked to identify who has thrown the javelin the farthest. Based on the test attempted by the students, the teacher concludes that 40% of the students have the misconception in the concept of decimal comparison and the rest do not have the misconception. 80% of the students having misconception answered Bijoy as the correct answer in the paper. 90% of the students who are identified with not having misconception, did not answer Bijoy as their answer. On the basis of the above information, answer the following questions : (i) What is the probability of a student not having misconception but still answers Bijoy in the test ? 1 (ii) What is the probability that a randomly selected student answers Bijoy as his answer in the test ? 1 (iii) (a) What is the probability that a student who answered as Bijoy is having misconception ? 2 OR (iii) (b) What is the probability that a student who answered as Bijoy is amongst students who do not have the misconception ? 2	Name of student	Distance of javelin (in meters)	Ajay	47.7	Bijoy	47.07	Kartik	43.09	Dinesh	43.9	Devesh	45.2
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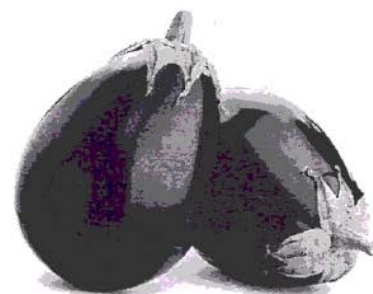
- 15.** A gardener wanted to plant vegetables in his garden. Hence he bought 10 seeds of brinjal plant, 12 seeds of cabbage plant and 8 seeds of radish plant. The shopkeeper assured him of germination probabilities of brinjal, cabbage and radish to be 25%, 35% and 40% respectively. But before he could plant the seeds, they got mixed up in the bag and he had to sow them randomly.



Radish



Cabbage



Brinjal

Based upon the above information, answer the following questions :

- (i) Calculate the probability of a randomly chosen seed to germinate. 2
- (ii) What is the probability that it is a cabbage seed, given that the chosen seed germinates ? 2

- 16.** A shop selling electronic items sells smartphones of only three reputed companies A, B and C because chances of their manufacturing a defective smartphone are only 5%, 4% and 2% respectively. In his inventory he has 25% smartphones from company A, 35% smartphones from company B and 40% smartphones from company C.

A person buys a smartphone from this shop.

- (i) Find the probability that it was defective. 2
- (ii) What is the probability that this defective smartphone was manufactured by company B ? 2

- 17.** Based upon the results of regular medical check-ups in a hospital, it was found that out of 1000 people, 700 were very healthy, 200 maintained average health and 100 had a poor health record.
- Let A_1 : People with good health,
 A_2 : People with average health,
 and A_3 : People with poor health.
- During a pandemic, the data expressed that the chances of people contracting the disease from category A_1 , A_2 and A_3 are 25%, 35% and 50%, respectively.
- Based upon the above information, answer the following questions :
- (i) A person was tested randomly. What is the probability that he/she has contracted the disease ? 2
- (ii) Given that the person has not contracted the disease, what is the probability that the person is from category A_2 ? 2

18.



Read the following passage and answer the questions given below.

There are two anti-aircraft guns, named as A and B. The probabilities that the shell fired from them hits an airplane are 0.3 and 0.2 respectively. Both of them fired one shell at an airplane at the same time.

- (i) What is the probability that the shell fired from exactly one of them hit the plane?
- (ii) If it is known that the shell fired from exactly one of them hit the plane, then what is the probability that it was fired from B?

19. Read the following passage and answer the questions given below:

In an Office three employees Jayant, Sonia and Oliver process incoming copies of a certain form. Jayant processes **50%** of the forms, Sonia processes **20%** and Oliver the remaining **30%** of the forms. Jayant has an error rate of **0.06**, Sonia has an error rate of **0.04** and Oliver has an error rate of **0.03**.

Based on the above information, answer the following questions.



- (i) Find the probability that Sonia processed the form and committed an error.
- (ii) Find the total probability of committing an error in processing the form.
- (iii) The manager of the Company wants to do a quality check. During inspection, he selects a form at random from the days output of processed form. If the form selected at random has an error, find the probability that the form is **not** processed by Jayant.

OR

- (iii) Let E be the event of committing an error in processing the form and let E_1, E_2 and E_3 be the events that Jayant, Sonia and Oliver processed the form. Find the value of $\sum_{i=1}^3 P(E_i|E)$.

20. Arka bought two cages of birds: Cage-I contains 5 parrots and 1 owl and Cage –II contains 6 parrots. One day Arka forgot to lock both cages and two birds flew from Cage-I to Cage-II (simultaneously). Then two birds flew back from cage-II to cage-I(simultaneously).

Assume that all the birds have equal chances of flying.

On the basis of the above information, answer the following questions:-

- (i) When two birds flew from Cage-I to Cage-II and two birds flew back from Cage-II to Cage-I then find the probability that the owl is still in Cage-I. **[2 Marks]**
- (ii) When two birds flew from Cage-I to Cage-II and two birds flew back from Cage-II to Cage-I, the owl is still seen in Cage-I, what is the probability that one parrot and the owl flew from Cage-I to Cage-II? **[2 Marks]**

21. Excessive use of screens can result in vision problems, obesity, sleep disorders, anxiety, low retention problems and can impede social and emotional comprehension and expression. It is essential to be mindful of the amount of time we spend on screens and to reduce our screen-time by taking regular breaks, setting time limits, and engaging in non-screen-based activities.



In a class of students of the age group 14 to 17, the students were categorised into three groups according to a feedback form filled by them. The first group constituted of the students who spent more than 4 hours per day on the mobile screen or the gaming screens, while the second group spent 2 to 4 hours /day on the same activities. The third group spent less than 2 hours /day on the same. The first group with the high screen time is 60% of all the students, whereas the second group with moderate screen time is 30% and the third group with low screen time is only 10% of the total number of students. It was observed that 80% students of first group faced severe anxiety and low retention issues, with 70% of second group, and 30% of third group having the same symptoms.

- I. What is the total percentage of students who suffer from anxiety and low retention issues in the class? [2]
- II. A student is selected at random, and he is found to suffer from anxiety and low retention issues. What is the probability that he/she spends screen time more than 4 hours per day? [2]